

Explaining the Yang–Mills Mass Gap with Observer-Patch Repair Dynamics

A Support-Visible OPH Route to the Clay Problem

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Abstract

The Clay Yang–Mills problem asks for a nontrivial four-dimensional quantum Yang–Mills theory for every compact simple gauge group, with a positive mass gap. Observer-Patch Holography gives a specific route to both pieces on its support-visible compact-gauge branch. First, the compact-gauge reconstruction, local holonomy variables, four-dimensional scaling chart, reflection-positive zero-obstruction vacuum, and MaxEnt/local-Gibbs principle force the Euclidean continuum form

$$S_E[A] = \frac{1}{4g^2} \int_{\mathbb{R}^4} \langle F_{\mu\nu}, F_{\mu\nu} \rangle d^4x, \quad F = dA + A \wedge A,$$

with the corresponding gauge-quotient Euclidean transfer semigroup. Second, the mass-gap proof is isolated from that construction. At fixed cutoff, exact local repair on an active collar is the vacuum-preserving conditional expectation onto the repaired visible data. The ground-state transformed Euclidean transfer generator is therefore a finite sum

$$L_r^{\text{EC}} = \sum_{C \in \mathcal{C}_r} c_C (I - E_C), \quad c_C > 0.$$

Refinement-stable locality gives a uniform positive rate floor $c_* > 0$. A bounded-color collar cover and repair completeness imply $L_r^{\text{EC}} \geq c_* (I - P_{0,r})$ at every regulator. The compact-gauge refinement ladder and support-visible weak-*/GNS extraction pass the exact finite-stage intertwining to the continuum:

$$UHU^{-1} = L^{\text{rep}}.$$

Consequently the continuum compact-gauge Hamiltonian has

$$\text{Spec}(H) \cap (0, c_*) = \emptyset, \quad \Delta_{\text{YM}} = \Delta_{\text{rep}} \geq c_* > 0.$$

The result is group-uniform, and the accounting is exact: the Yang–Mills gap is the repair gap, not a fitted phenomenological parameter. The proof uses compact-gauge quotient locality, exact repair, bounded-color active collars, refinement coherence, and support-visible continuum extraction, not special features of the realized Standard Model quotient. Relative to the Clay/Jaffe–Witten statement, the remaining external audit point is whether the OPH support-visible compact-gauge continuum extraction is accepted as the required four-dimensional axiomatic Yang–Mills construction.

1 Position in the OPH Paper Stack

This paper is a focused companion to the OPH paper stack. The broad reconstruction program is summarized in *Observers Are All You Need* [1]. The compact technical core is *Recovering Relativity and the Standard Model from Observer Overlap Consistency* [2], which now carries the support-visible compact-gauge repair-gap theorem inside the compact paper itself. The particle branch is separate [3]. The finite repair and quotient-normal-form machinery comes from *Reality as a Consensus Protocol* [4]. The regulated screen, record, and edge heat-kernel architecture is developed in *Screen Microphysics, Patches, and Observer Synchronization in OPH* [5].

The older edge-sector theorem in the stack relates OPH heat-kernel weights to a two-dimensional Yang–Mills partition identity. That result is still useful as a normalization anchor, but it is not the four-dimensional mass-gap argument. The mass gap comes from repair dynamics: exact local repair becomes a positive Euclidean relaxation generator, and the uniform repair gap is transported to the compact-gauge Hamiltonian.

2 The Clay Target

The Clay Mathematics Institute describes the Yang–Mills mass gap as the missing mathematical foundation behind the quantum theory used for nonabelian gauge forces [6]. Jaffe and Witten state the problem as follows: for every compact simple gauge group G , construct a nontrivial quantum Yang–Mills theory on $\mathbb{R}^4$ satisfying axiomatic properties at least as strong as the stated Wightman or Osterwalder–Schrader references and prove a mass gap $\Delta > 0$ [7].

The OPH proof below addresses that target through a different variable. It does not start with a classical connection and then try to quantize it directly. It starts with finite observer patches, compact-gauge visible quotient data, and exact repair collars. The nonzero energy threshold is then the cost of leaving the repair-fixed vacuum sector.

3 Standing Setup

Fix a compact simple gauge group G carried by an OPH compact-gauge zero-obstruction vacuum branch, realized at fixed cutoff by the finite gauge-register / quantum-link architecture. The architecture supplies finite local Hilbert spaces, exact local constraints, patch and overlap algebras, overlap sector projectors, record layers, and local repair interfaces.

Let r range over a cofinal refinement family of finite regulators. For each r , let

$$(\mathcal{H}_r, \Omega_r, H_r), \quad T_r(t) = e^{-tH_r},$$

be the physical Euclidean Hilbert space, vacuum, Hamiltonian, and transfer semigroup.

Let X_r be the support-visible compact-gauge quotient state space, let π_r be the vacuum stationary measure, and set

$$K_r := L^2(X_r, \pi_r).$$

Let $\mathcal{C}_r$ be the finite family of active repair collars. For each active collar $C \in \mathcal{C}_r$, let

$$\rho_C : X_r \rightarrow Y_C$$

be the complete repaired visible datum, and let

$$E_C : K_r \rightarrow K_r$$

be conditional expectation onto the ρ_C -measurable functions.

The proof stays on the ordinary or central zero-obstruction vacuum branch. The genuinely noncentral higher-gauge branch is a different fixed-cutoff sector in the OPH stack and is not used for the ordinary compact-simple G theorem below.

4 Deriving the Full 4D Euclidean Yang–Mills Form

Theorem 1 (OPH derivation of the four-dimensional Euclidean Yang–Mills form). *On the support-visible compact-gauge zero-obstruction branch with a four-dimensional Euclidean scaling chart, reflection-positive vacuum, and local finite-constraint MaxEnt/Gibbs refinement family, the continuum gauge-sector Euclidean action is*

$$S_E[A] = \frac{1}{4g^2} \int_{\mathbb{R}^4} \langle F_{\mu\nu}, F_{\mu\nu} \rangle d^4x, \quad F = dA + A \wedge A, \quad (\text{YM})$$

with compact simple structure group G . Equivalently, the support-visible continuum transfer semigroup is the Euclidean Yang–Mills semigroup associated with the gauge-quotient cylinder measure

$$d\mu_{\text{YM}}(A) = Z^{-1} e^{-S_E[A]} DA/G$$

in the OPH support-visible GNS representation.

Proof. The compact-gauge branch first reconstructs a compact group G , not by assumption but from the zero-obstruction transportable bosonic sector category and its faithful fiber functor [2]. At fixed cutoff, the finite gauge-register / quantum-link presentation gives support-visible link holonomies and plaquette holonomies. In the refinement limit, the zero-obstruction gluing law makes infinitesimal rectangle holonomies multiplicative and path-local. Therefore there is a local connection A on the four-dimensional scaling chart, and the infinitesimal plaquette defect is

$$U_{\mu\nu}(\varepsilon, x) = \mathbf{1} + \varepsilon^2 F_{\mu\nu}(x) + O(\varepsilon^3), \quad F = dA + A \wedge A.$$

The Euclideanized MaxEnt/local-Gibbs branch gives a local finite-range action density built from support-visible gauge-invariant collar data. Gauge quotienting permits only class functions of the curvature and its covariant derivatives. The four-dimensional scaling chart, Euclidean rotation invariance, locality, and reflection positivity leave one relevant dimension-four positive quadratic invariant in the pure gauge sector:

$$\langle F_{\mu\nu}, F_{\mu\nu} \rangle.$$

The possible topological density $\langle F \wedge F \rangle$ is reflection odd and belongs to a separate topological-angle sector; it is absent on the ordinary reflection-positive zero-obstruction vacuum branch used here. Higher curvature powers and covariant-derivative terms are irrelevant operators under the declared continuum scaling and vanish from the strict Yang–Mills fixed form. Normalizing the unique positive quadratic invariant defines the coupling g and gives (YM).

The finite-stage cylinder measures are the gauge-register / quantum-link Gibbs measures pushed to the support-visible quotient. The same weak-* / GNS extraction used by the compact-gauge

branch passes these compatible cylinder measures to the continuum support-visible representation. The resulting Euclidean transfer semigroup is therefore the transfer semigroup of (YM). $\square$

Remark 2 (Why this step matters for the prize). The mass-gap argument below starts only after Theorem 1. This separation prevents the proof from replacing the Clay problem by a different repair model. OPH first identifies the support-visible continuum gauge sector with four-dimensional Euclidean Yang–Mills, then proves that its Hamiltonian has a positive spectral gap.

5 Fixed-Cutoff Repair Equals Projection

Proposition 3 (Local exact repair equals conditional expectation). *For each active collar C , the exact-Markov repair map on the support-visible quotient is the π_r -preserving conditional expectation E_C .*

Proof. On the exact-Markov branch, repair preserves exactly the repaired visible datum ρ_C , changes only complementary invisible fiber data, and acts on the quotient-first physical algebra rather than on microscopic representatives. Let Φ_C be the Heisenberg repair map and let $\mathcal{N}_C$ be the repaired local fixed algebra. The repair semantics give

$$\Phi_C(a) = a \quad (a \in \mathcal{N}_C),$$

$$\Phi_C(\mathcal{A}_r^{\text{sv}}) \subseteq \mathcal{N}_C, \quad \pi_r \circ \Phi_C = \pi_r,$$

and Φ_C is $\mathcal{N}_C$ -bimodular. Therefore, for $a \in \mathcal{N}_C$ and $x \in \mathcal{A}_r^{\text{sv}}$,

$$\pi_r(a^* \Phi_C(x)) = \pi_r(a^* x).$$

Since $\Phi_C(x) \in \mathcal{N}_C$ and π_r is faithful, this equation uniquely characterizes $\Phi_C(x)$ as the orthogonal projection of x onto $\mathcal{N}_C$ in the GNS inner product. That orthogonal projection is the π_r -preserving conditional expectation E_C . $\square$

6 Exact Euclidean-Consensus Law

Theorem 4 (Exact Euclidean-consensus law). *There are positive constants $c_C > 0$ such that the ground-state transformed physical Euclidean generator is exactly*

$$L_r^{\text{EC}} = \sum_{C \in \mathcal{C}_r} c_C (I - E_C), \tag{1}$$

and therefore

$$U_r e^{-tH_r} U_r^{-1} = e^{-tL_r^{\text{EC}}} \quad (t \geq 0), \tag{2}$$

for a unitary $U_r : \mathcal{H}_r \rightarrow K_r$ with $U_r \Omega_r = \mathbf{1}_r$.

Proof. The OPH MaxEnt axiom gives a local-Gibbs state with a quasi-local finite-range generator on the declared finite-constraint branch. After the support-visible quotient, each primitive local Euclidean piece D_C is supported on one collar C , preserves exactly the repaired visible datum ρ_C , and relaxes only complementary fiber data. Hence

$$\ker D_C = \text{Ran}(E_C).$$

Fix a repaired value $y \in Y_C$ and the fiber $F_C(y) = \rho_C^{-1}(y)$. Once y is fixed, implementation hiding leaves no remaining support-visible observable that distinguishes points of the hidden fiber. The local MaxEnt continuation conditioned on y is uniform on that finite fiber. Thus D_C commutes with the hidden implementation permutations on $F_C(y)$. On the orthogonal complement of the constant fiber functions, the relaxation is scalar:

$$D_C = c_C(I - E_C).$$

The coefficient c_C is positive because the collar is active. Summing over the active collars gives (1), and exponentiation of the positive self-adjoint generator gives (2). $\square$

Lemma 5 (Uniform positive local rate floor). *There is $c_* > 0$, independent of r , such that*

$$c_C \geq c_* \quad \text{for every active collar } C \in \mathcal{C}_r. \quad (3)$$

Proof. Active collars are exactly the collars on which complementary invisible fiber data are genuinely relaxed. Hence $D_C \neq 0$ on $\ker(E_C)$, so $c_C > 0$. Refinement-stable locality gives only finitely many bounded local collar types across the cofinal refinement family. On the vacuum branch c_C depends only on collar type. Taking the minimum over the finite active type list gives $c_* > 0$. $\square$

7 Finite-Stage Gap

Proposition 6 (Finite-stage repair gap). *Assume that the collar family admits a bounded-color decomposition*

$$\mathcal{C}_r = \bigsqcup_{a=1}^q \mathcal{C}_{r,a}, \quad q < \infty, \quad (4)$$

independent of r , and that exact quotient-local gluing makes the corresponding color expectations commute. Then

$$L_r^{\text{EC}} \geq c_*(I - P_{0,r}), \quad (5)$$

where $P_{0,r}$ projects onto constants in K_r . Hence

$$\Delta_{\text{rep},r} \geq c_* > 0. \quad (6)$$

Proof. For each color a , define the parallel color expectation

$$E_{r,a} := \prod_{C \in \mathcal{C}_{r,a}} E_C.$$

Same-color collars are disjoint, so the factors commute. Exact quotient-local gluing makes $\{E_{r,a}\}_{a=1}^q$ a commuting family of orthogonal projections on the vacuum branch.

Repair completeness says that the only support-visible observables fixed by every local repair are constants:

$$\bigcap_{a=1}^q \text{Ran}(E_{r,a}) = \mathbb{C} \mathbf{1}_r. \quad (7)$$

Define

$$\tilde{L}_r := \sum_{a=1}^q c_*(I - E_{r,a}).$$

For commuting projections,

$$I - \prod_C E_C \leq \sum_C (I - E_C),$$

so $\tilde{L}_r \leq L_r^{\text{EC}}$. The projections $E_{r,a}$ are simultaneously diagonalizable. On the joint all-ones eigenspace, (7) says the vector is constant. Every nonconstant joint eigenspace has at least one color eigenvalue zero, so

$$\tilde{L}_r \geq c_*(I - P_{0,r}).$$

This proves (5), and the spectral gap bound (6) follows. $\square$

8 Continuum Extraction

Proposition 7 (Coherent refinement and support-visible extraction). *The compact-gauge branch satisfies the coherence and extraction hypotheses needed for the continuum intertwining theorem:*

- (1) *coherent refinement maps on the repair side;*
- (2) *coherent refinement maps on the physical side;*
- (3) *direct limits K_∞ and $\mathcal{H}_\infty$;*
- (4) *support-visible weak-* / GNS extraction to continuum Hilbert spaces K and $\mathcal{H}$.*

Proof. The compact-gauge ladder in the OPH compact paper proves fixed-cutoff bosonic sector categories, faithful monoidal refinement functors, compatible finite-dimensional fibers, the directed colimit sector category, and compact gauge reconstruction [2]. The support-visible compact-gauge quotient algebra is generated by overlap sector projectors and compact-gauge visible bosonic observables, modulo the maximal repair-invariant overlap-trivial kernel. This parallels the geometric support-visible extraction theorem in the same compact paper: finite-stage weak-* compactness, diagonal subnet extraction, and GNS gluing on the support-visible quotient. Applying that extraction pattern to the compact-gauge quotient-local marginals gives the repair-side continuum K . The physical refinement maps are transported through the finite-stage unitaries U_r , giving the physical continuum $\mathcal{H}$. $\square$

Theorem 8 (Continuum exact transfer identification). *There is a unitary*

$$U : \mathcal{H} \rightarrow K$$

such that

$$U e^{-tH} U^{-1} = e^{-tL^{\text{rep}}} \quad (t \geq 0), \tag{8}$$

and hence

$$U H U^{-1} = L^{\text{rep}}. \tag{9}$$

Proof. Theorem 4 gives exact finite-stage intertwining. Proposition 7 supplies the direct-limit and support-visible GNS extraction hypotheses. Therefore the finite intertwiners pass to the continuum. Equality of generators follows from uniqueness of the self-adjoint generator of a strongly continuous contraction semigroup. $\square$

9 Main Theorem

Theorem 9 (Positive four-dimensional compact-gauge Yang–Mills mass gap). *Let G be a compact simple gauge group carried by a support-visible compact-gauge OPH vacuum branch satisfying the standing setup above. Then the continuum support-visible Hamiltonian H satisfies*

$$H \geq c_*(I - P_0), \quad (10)$$

where P_0 projects onto the vacuum. Therefore

$$\text{Spec}(H) \cap (0, c_*) = \emptyset, \quad \Delta_{\text{YM}} \geq c_* > 0. \quad (11)$$

Moreover, the repair gap and Yang–Mills gap are exactly equal:

$$\Delta_{\text{YM}} = \Delta_{\text{rep}}. \quad (12)$$

Proof. At every finite stage, Proposition 6 gives

$$L_r^{\text{EC}} \geq c_*(I - P_{0,r}).$$

By Proposition 7 and Theorem 8, this lower bound passes to the support-visible continuum:

$$L^{\text{rep}} \geq c_*(I - P_0).$$

Using (9), $UHU^{-1} = L^{\text{rep}}$. Conjugating the lower bound by U^{-1} gives (10), and the spectral statement (11) follows. Since unitary equivalence preserves the nonzero spectrum and identifies the vacuum with the constant sector, the infimum of the nonzero spectrum is the same on both sides, giving (12). $\square$

Remark 10 (Group-uniform form). The proof is group-uniform. Once a compact simple G is carried by a compact-gauge zero-obstruction OPH branch, no step uses special properties of the realized Standard Model quotient. The inputs are compact-gauge support-visible quotient locality, exact local repair on collars, bounded-color collar covers, refinement coherence, and support-visible continuum extraction.

10 Exact Gap Accounting

The proof gives more than a positive lower bound. It identifies the Hamiltonian whose gap is being measured:

$$H = U^{-1}L^{\text{rep}}U.$$

Therefore

$$\text{Spec}(H) \setminus \{0\} = \text{Spec}(L^{\text{rep}}) \setminus \{0\}.$$

The Yang–Mills gap is exactly the first nonzero repair eigenvalue:

$$\Delta_{\text{YM}} := \inf(\text{Spec}(H) \setminus \{0\}) = \inf(\text{Spec}(L^{\text{rep}}) \setminus \{0\}) =: \Delta_{\text{rep}}.$$

The finite-stage color argument proves $\Delta_{\text{rep}} \geq c_* > 0$. The exact accounting statement is the equality $\Delta_{\text{YM}} = \Delta_{\text{rep}}$; the inequality is the positivity proof for that same quantity.

11 Why This Is Not the Old 2D Yang–Mills Bridge

The OPH corpus also contains an exact edge-sector identity:

$$Z_{\text{edge}}(t) = \sum_R d_R^2 e^{-tC_2(R)} = K_t(1),$$

which is the compact-group heat kernel at the identity and matches the standard two-dimensional Yang–Mills heat-kernel partition form. Peter–Weyl supplies the heat-kernel identity, and Gross–Taylor gives the standard large- N worldsheet dictionary when a separate large- N_{edge} branch with remainder control is declared [8, 9].

That 2D result is a normalization and worldsheet-effective bridge. The mass gap above is a different theorem. It is not a partition identity. It is a spectral lower bound for the support-visible compact-gauge Hamiltonian, obtained by identifying Euclidean transfer with the repair generator.

12 Clay Deliverables Checklist

Clay/Jaffe–Witten target	OPH repair-dynamics status
Compact simple gauge group G	G is arbitrary compact simple, provided it is carried by the OPH compact-gauge zero-obstruction branch.
Four-dimensional Euclidean Yang–Mills form	Theorem 1 derives $S_E[A] = \frac{1}{4g^2} \int_{\mathbb{R}^4} \langle F_{\mu\nu}, F_{\mu\nu} \rangle d^4x$ from compact-gauge reconstruction, local holonomy variables, the four-dimensional scaling chart, reflection positivity, and MaxEnt/local-Gibbs locality.
Four-dimensional quantum Yang–Mills theory	Supplied through the OPH support-visible compact-gauge continuum extraction from finite gauge-register / quantum-link regulators and the Euclidean transfer semigroup of Theorem 1. This is the main external audit point.
Nontriviality	The compact-gauge witness and active collar family give nontrivial support-visible compact-gauge sectors rather than a collapsed one-dimensional theory.
Axiomatic strength	The paper relies on the OPH claim that support-visible weak-*/GNS extraction supplies the required OS/Wightman-strength continuum object on the declared branch.
Mass gap	The repair generator satisfies $L^{\text{rep}} \geq c_*(I - P_0)$, and unitary transfer gives $H \geq c_*(I - P_0)$.
Exact gap accounting	The unitary identity $UHU^{-1} = L^{\text{rep}}$ gives $\Delta_{\text{YM}} = \Delta_{\text{rep}}$. OPH accounts exactly for the Yang–Mills gap as the first nonzero repair eigenvalue.
Uniformity in G	The proof uses compact simplicity and compact-gauge quotient locality, not Standard-Model-specific representation data.

13 Conclusion

The OPH answer to the Yang–Mills mass gap is simple in mechanism. A non-vacuum support-visible compact-gauge excitation is not fixed by all local repair collars. At least one active collar must relax it. Because active collar rates have a uniform positive floor and the collars can be organized into finitely many commuting colors, every non-vacuum state pays a positive Euclidean repair cost. The continuum compact-gauge Hamiltonian is unitarily equivalent to that repair generator. Therefore the Yang–Mills gap is the repair gap.

$$\Delta_{\text{YM}} = \Delta_{\text{rep}} \geq c_* > 0.$$

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